

AAE 50700: Principles of Dynamics
Problem Set 5 Report

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(5.1) Orbital Dynamics with J_2 and Atmospheric Drag

(a): Given

Consider a spacecraft of mass m orbiting Earth perturbed by J_2 and atmosphere drag. Let the inertial frame be $\mathcal{N} : \{\hat{e}_x, \hat{e}_y, \hat{e}_z\}$ and write its Cartesian position as $\mathbf{r} = x\hat{e}_x + y\hat{e}_y + z\hat{e}_z, r = \|\mathbf{r}\|_2$. Expressing the potential energy for Earth gravity under J_2 in the Cartesian coordinates, we have

$$V(x, y, z) = -\frac{\mu m}{r} + \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2), \quad (1)$$

where μ is Earth's gravitational parameter and $r_o = R_{\text{earth}}$ is its equatorial radius.

We use the same model as considered in PS1 to model the drag, which assumes an inertially stationary, spherically symmetric atmosphere, given by

$$\mathbf{F}_D = m\mathbf{a}_D, \quad \mathbf{a}_D = -\frac{1}{2}C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}}, \quad v := \|\mathbf{v}\|_2, \quad \hat{\mathbf{v}} := \frac{\mathbf{v}}{v}, \quad (2)$$

where $\mathbf{v} = \dot{\mathbf{r}}$, C_D is the drag coefficient, A is the reference area, and

$$\rho(r) \approx \rho_0 \exp \left[-\frac{r - 6978 \text{ km}}{H} \right]. \quad (3)$$

(a.1): Find

From the Lagrangian using the Cartesian coordinates as generalized coordinates, $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

First recall that the kinetic energy T of an object in motion can be expressed by half the mass times the square of the velocity:

$$T(\mathbf{v}) = \frac{m}{2} \mathbf{v}^T \mathbf{v} = \frac{mv^2}{2} \quad (4)$$

Then, recall the Lagrangian L is the kinetic energy minus the potential energy $L = T - V$ (V given by the problem definition):

$$L(\mathbf{r}, \mathbf{v}) = \frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \quad (5)$$

(a.2): Find

Derive the generalized force for Eq. (2) associated with $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

It is known that virtual work δw is translational force $\mathbf{F}$ dotted with the Jacobian of position with respect to the generalized coordinates multiplied by the displacement of generalized coordinates. This also equals generalized force $\mathbf{Q}$ dotted with the same displacement:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \delta \mathbf{q} = \mathbf{Q}^T \delta \mathbf{q}. \quad (6)$$

Since in this case the generalized coordinates are the Cartesian position, $\mathbf{q} = \mathbf{r}$, the Jacobian is the identity matrix, and the generalized force is simply the applied force:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \delta \mathbf{r} = \mathbf{F}^T I \delta \mathbf{r} = \mathbf{F}^T \delta \mathbf{r} \Rightarrow \mathbf{Q} = \mathbf{F}. \quad (7)$$

So, from Eq. (2) there is only one force, which is the force of drag $\mathbf{F}_D$. This means the generalized force from that equation associated with $\mathbf{q}$ is $\mathbf{Q} = \mathbf{F}_D$. The mass terms cancel and simplify to get generalized force:

$$\mathbf{Q} = \mathbf{F}_D = m \mathbf{a}_D = m \left(-\frac{1}{2} C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}} \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \quad (8)$$

(a.3): Find

Derive the Lagrange's equations and write the result as the Cartesian vector EOM

$$\ddot{\mathbf{r}} = \mathbf{a}_{2B} + \mathbf{a}_{J2} + \mathbf{a}_D \quad (9)$$

Analysis/Conclusions:

We see here that this system has three degrees of freedom with no redundancy in coordinates, so $\mathbf{q}$ is a minimal coordinate set, and the system has no constraints on movement. Since there is also a non-conservative force (drag), the general form of Lagrange's equations with non-conservative general force may be used:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}_{nc} \quad (10)$$

And since, $\mathbf{q} = \mathbf{r}$ and $\mathbf{F}_D$ is the only non-conservative general force:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{v}}} \right) - \frac{\partial L}{\partial \mathbf{r}} = \mathbf{F}_D \quad (11)$$

Then we compute the derivatives in Eq. (11) individually.

So first to solve $\frac{\partial L}{\partial \mathbf{v}}$:

$$\begin{aligned}\frac{\partial L}{\partial \mathbf{v}} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} \frac{\partial v^2}{\partial \mathbf{v}} + \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m}{r} \right) - \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} 2\mathbf{v}^T + 0 - 0 = m\mathbf{v}^T\end{aligned}\tag{12}$$

Then to solve $\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right)$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = m \frac{d\mathbf{v}^T}{dt} = m\mathbf{a}^T\tag{13}$$

Then solve $-\frac{\partial L}{\partial \mathbf{r}}$:

$$\begin{aligned}-\frac{\partial L}{\partial \mathbf{r}} &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} \right) - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= 0 - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right)\end{aligned}\tag{14}$$

Then for readability, we do the remaining component derivatives individually:

$$\frac{\partial r^{-1}}{\partial \mathbf{r}} = \frac{\partial r^{-1}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} = -r^{-2} \hat{\mathbf{r}}^T\tag{15}$$

And the other component:

$$\begin{aligned}
&= \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(\frac{\partial}{\partial \mathbf{r}} (3r^{-5} z^2) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \frac{\partial z}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \hat{\mathbf{e}}_z^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(\hat{\mathbf{r}}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4} \hat{\mathbf{r}}^T) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(r^{-1} \mathbf{r}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4})(r^{-1} \mathbf{r}^T) \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 \left(-5r^{-7} z^2 \mathbf{r}^T + 2r^{-5} z \hat{\mathbf{e}}_z^T + r^{-5} \mathbf{r}^T \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 r^{-5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{16}$$

So then that gives:

$$\begin{aligned}
-\frac{\partial L}{\partial \mathbf{r}} &= -\mu m \left(-r^{-2} \hat{\mathbf{r}}^T \right) + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{17}$$

Which then plugging the above terms back into Eq. (11) gives:

$$m\mathbf{a}^T + \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}}^T \tag{18}$$

Which then solved for acceleration $\mathbf{a} = \ddot{\mathbf{r}}$:

$$\begin{aligned}
m\mathbf{a} &= -\frac{\mu m}{r^3} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu m}{r^3 m} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5 m} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu}{r^3} \mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}}
\end{aligned} \tag{19}$$

Which means the Cartesian vector EOM are:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) - \frac{1}{2m}C_D A \rho(r)v^2\hat{\mathbf{v}} \quad (20)$$

Where:

$$\begin{aligned} \mathbf{a}_{2B} &= -\frac{\mu}{r^3}\mathbf{r} \\ \mathbf{a}_{J2} &= -\frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) \\ \mathbf{a}_D &= -\frac{1}{2m}C_D A \rho(r)v^2\hat{\mathbf{v}} \end{aligned} \quad (21)$$

Since these terms come from the two-body problem, J_2 perturbation, and drag acceleration components of the system respectively.

(b): Given

Now consider spherical coordinates $\mathbf{q} = [r, \phi, \theta]^T$, where r is radius, ϕ is azimuth about $\hat{\mathbf{e}}_z$, and θ is elevation measured from $\hat{\mathbf{e}}_x - \hat{\mathbf{e}}_y$ plane. The potential energy for gravity with J_2 is given by:

$$V = -\frac{m\mu}{r} \left[1 - J_2 \left(\frac{r_o}{r} \right)^2 P_2(\sin \theta) \right] \quad (22)$$

(b.1): Find

Form the Lagrangian using spherical coordinates.

Analysis/Conclusions:

The Lagrangian is once again found by kinetic minus potential energy $L = T - V$. The gravity potential including J_2 zonal harmonics was already given by Eq. (22), then the kinetic energy is once again given by:

$$T = \frac{mv^2}{2} \quad (23)$$

To express T in spherical coordinates, we write the position as $\mathbf{r} = r\hat{\mathbf{e}}_r$ in the spherical basis $\mathcal{S} : \{\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_\theta\}$. Since this basis rotates relative to the inertial basis $\mathcal{N} : \{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$, we find $\mathbf{v}$ with the transport theorem, which requires the rotation of the spherical basis with respect to the inertial Cartesian basis, $\boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}}$. We can represent this by considering $\dot{\phi}$ scaled by the $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ components of $\hat{\mathbf{e}}_z = \sin \theta \hat{\mathbf{e}}_r + \cos \theta \hat{\mathbf{e}}_\theta$, as well as $\dot{\theta}$ scaled by the negative direction of $\hat{\mathbf{e}}_\phi$:

$$\boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} = \dot{\phi} \sin \theta \hat{\mathbf{e}}_r + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\theta - \dot{\theta} \hat{\mathbf{e}}_\phi \quad (24)$$

Which we then find $\mathbf{v}$ via the transport theorem as:

$$\begin{aligned} \mathbf{v} &= \frac{{}^S d}{dt} \mathbf{r} + \boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} \times \mathbf{r} \\ &= \frac{{}^S d}{dt} (r\hat{\mathbf{e}}_r) + \boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} \times (r\hat{\mathbf{e}}_r) \\ &= \frac{{}^S dr}{dt} \hat{\mathbf{e}}_r + r \frac{{}^S d\hat{\mathbf{e}}_r}{dt} + r \left(\dot{\phi} \sin \theta \hat{\mathbf{e}}_r + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\theta - \dot{\theta} \hat{\mathbf{e}}_\phi \right) \times \hat{\mathbf{e}}_r \\ &= \dot{r} \hat{\mathbf{e}}_r + r(0) + r \left(\dot{\phi} \sin \theta (\mathbf{0}) + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\phi - \dot{\theta} (-\hat{\mathbf{e}}_\theta) \right) \\ &= \dot{r} \hat{\mathbf{e}}_r + r \cos \theta \dot{\phi} \hat{\mathbf{e}}_\phi + r \dot{\theta} \hat{\mathbf{e}}_\theta \end{aligned} \quad (25)$$

Then we can simply use the $\mathcal{S}$ -frame representation of $\mathbf{v}$ to find an expression

for the kinetic energy T :

$$\begin{aligned}
T &= \frac{mv^2}{2} = \frac{m}{2} \mathbf{v}^T \mathbf{v} = \frac{m}{2} (v_r^2 + v_\phi^2 + v_\theta^2) \\
&= \frac{m}{2} \left(\dot{r}^2 + (r \cos \theta \dot{\phi})^2 + (r \dot{\theta})^2 \right) \\
&= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right)
\end{aligned} \tag{26}$$

So, the Lagrangian can be written as:

$$L = \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) + \frac{m\mu}{r} \left[1 - J_2 \left(\frac{r_o}{r} \right)^2 P_2(\sin \theta) \right] \tag{27}$$

However, this expression still needs to have substituted in the second degree, zeroth order, Legendre polynomial $P_2(\sin \theta)$, which can be found by the following formula:

$$P_n(x) = \sum_{m=0}^M (-1)^m \frac{(2n-2m)!}{2^n m! (n-m)! (n-2m)!} x^{n-2m} \tag{28}$$

Where n is degree and $M = n/2$.

Evaluating it becomes:

$$\begin{aligned}
P_2(\sin \theta) &= \sum_{m=0}^1 (-1)^m \frac{(2(2)-2m)!}{2^2 m! (2-m)! (2-2m)!} \sin^{2-2m} \theta \\
&= (-1)^0 \frac{(4-0)!}{2^2 (0)! (2-0)! (2-0)!} \sin^2 \theta + (-1)^1 \frac{(4-2)!}{2^2 (1)! (2-1)! (2-2)!} \sin^0 \theta \\
&= \frac{24}{4 \cdot 1 \cdot 2 \cdot 2} \sin^2 \theta - \frac{2}{4 \cdot 1 \cdot 1 \cdot 1} \\
&= \frac{3}{2} \sin^2 \theta - \frac{1}{2}
\end{aligned} \tag{29}$$

So then substituting into the found Lagrangian, we get its proper form:

$$\begin{aligned}
L &= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) + \frac{m\mu}{r} \left(1 - J_2 \left(\frac{r_o}{r} \right)^2 \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{m\mu J_2 r_o^2}{r^3} \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3}
\end{aligned} \tag{30}$$

(b.2): Find

Derive the generalized force for Eq. (2) associated with $\mathbf{q} = [r, \phi, \theta]^T$.

Analysis/Conclusions:

In part (a.2), $\mathbf{q} = \mathbf{r}$ made the Jacobian in $\mathbf{Q} = \mathbf{F}_D^T \frac{\partial \mathbf{r}}{\partial \mathbf{q}}$ the identity matrix, but now that $\mathbf{q} = [r, \phi, \theta]^T$, we must find how the position changes with respect to r , ϕ , and θ .

Let us consider the total derivative of $\mathbf{r}$ from the perspective of the chain rule:

$$\frac{d\mathbf{r}}{dt} = \frac{\partial \mathbf{r}}{\partial r} \dot{r} + \frac{\partial \mathbf{r}}{\partial \phi} \dot{\phi} + \frac{\partial \mathbf{r}}{\partial \theta} \dot{\theta} = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \dot{\mathbf{q}} \quad (31)$$

Then put it side by side with the expression found for $\mathbf{v} = \frac{d\mathbf{r}}{dt}$:

$$\mathbf{v} = \hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \quad (32)$$

Thus from this we see that:

$$\frac{\partial \mathbf{r}}{\partial r} = \hat{\mathbf{e}}_r, \quad \frac{\partial \mathbf{r}}{\partial \phi} = r \cos \theta \hat{\mathbf{e}}_\phi, \quad \frac{\partial \mathbf{r}}{\partial \theta} = r \hat{\mathbf{e}}_\theta \quad (33)$$

So when we compute the generalized force using $Q_j = \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial q_j}$

$$\begin{aligned} Q_r &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial r} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot \hat{\mathbf{e}}_r \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot \hat{\mathbf{e}}_r \\ &= -\frac{1}{2} C_D A \rho(r) v \dot{r} \\ Q_\phi &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial \phi} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot (r \cos \theta \hat{\mathbf{e}}_\phi) \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot (r \cos \theta \hat{\mathbf{e}}_\phi) \\ &= -\frac{1}{2} C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \\ Q_\theta &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial \theta} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot (r \hat{\mathbf{e}}_\theta) \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot (r \hat{\mathbf{e}}_\theta) \\ &= -\frac{1}{2} C_D A \rho(r) v r^2 \dot{\theta} \end{aligned} \quad (34)$$

Which makes the generalized force:

$$\begin{aligned} Q_r &= -\frac{1}{2}C_D A \rho(r) v \dot{r} \\ Q_\phi &= -\frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \\ Q_\theta &= -\frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta} \end{aligned} \quad (35)$$

(b.3): Find

Derive the Lagrange's equations and write the result as $\ddot{r}=?$, $\ddot{\phi}=?$, and $\ddot{\theta}=?$.

Analysis/Conclusions:

Once again this system has a minimal coordinate set and a non-conservative force from drag. So, the general form of Lagrange's equations with non-conservative general force may be used:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}_{nc} \quad (36)$$

First, we evaluate the generalized momentum terms:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\phi}} &= \frac{\partial}{\partial \dot{\phi}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= 0 + \frac{mr^2 \cos^2 \theta}{2} \frac{\partial \dot{\phi}^2}{\partial \dot{\phi}} + 0 + 0 - 0 + 0 = mr^2 \dot{\phi} \cos^2 \theta \end{aligned} \quad (37)$$

And the θ momentum:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}} &= \frac{\partial}{\partial \dot{\theta}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= 0 + 0 + \frac{mr^2}{2} \frac{\partial \dot{\theta}^2}{\partial \dot{\theta}} + 0 - 0 + 0 = mr^2 \dot{\theta} \end{aligned} \quad (38)$$

And the r momentum:

$$\begin{aligned} \frac{\partial L}{\partial \dot{r}} &= \frac{\partial}{\partial \dot{r}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= \frac{m}{2} \frac{\partial \dot{r}^2}{\partial \dot{r}} + 0 + 0 + 0 - 0 + 0 = m\dot{r} \end{aligned} \quad (39)$$

Then we compute their time derivatives:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) &= \frac{d}{dt} (mr^2 \dot{\phi} \cos^2 \theta) \\
&= m \left(\dot{\phi} \cos^2 \theta \frac{dr^2}{dt} + r^2 \cos^2 \theta \frac{d\dot{\phi}}{dt} + r^2 \dot{\phi} \frac{d}{dt} \cos^2 \theta \right) \\
&= m \left(\dot{\phi} \cos^2 \theta \frac{dr^2}{dr} \frac{dr}{dt} + r^2 \cos^2 \theta \frac{d\dot{\phi}}{dt} + r^2 \dot{\phi} \frac{d}{d\theta} (\cos^2 \theta) \frac{d\theta}{dt} \right) \quad (40) \\
&= m \left(2r\dot{r}\dot{\phi} \cos^2 \theta + r^2 \ddot{\phi} \cos^2 \theta - 2r^2 \dot{\phi} \dot{\theta} \cos \theta \sin \theta \right) \\
&= mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right)
\end{aligned}$$

And the θ time derivative:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{d}{dt} (mr^2 \dot{\theta}) = m \left(\frac{dr^2}{dr} \frac{dr}{dt} \dot{\theta} + r^2 \frac{d\dot{\theta}}{dt} \right) = 2mr\dot{r}\dot{\theta} + mr^2 \ddot{\theta} = mr (2\dot{r}\dot{\theta} + r\ddot{\theta}) \quad (41)$$

And the r time derivative:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = m \frac{d\dot{r}}{dt} = m\ddot{r} \quad (42)$$

Since the Lagrangian is not a function of ϕ :

$$\frac{\partial L}{\partial \phi} = 0 \quad (43)$$

Then compute the θ derivative:

$$\begin{aligned}
\frac{\partial L}{\partial \theta} &= \frac{\partial}{\partial \theta} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= 0 + \frac{mr^2 \dot{\phi}^2}{2} \frac{\partial}{\partial \theta} (\cos^2 \theta) + 0 + 0 - \frac{3m\mu J_2 r_o^2}{2r^3} \frac{\partial}{\partial \theta} (\sin^2 \theta) + 0 \\
&= -mr^2 \dot{\phi}^2 \cos \theta \sin \theta - 3m\mu J_2 r_o^2 r^{-3} \cos \theta \sin \theta \\
&= -m \cos \theta \sin \theta \left(r^2 \dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^3} \right) \quad (44)
\end{aligned}$$

And the r derivative:

$$\begin{aligned}
&= \frac{\partial L}{\partial r} \\
&= \frac{\partial}{\partial r} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= 0 + \frac{m \cos^2 \theta \dot{\phi}^2}{2} \frac{dr^2}{dr} + \frac{m\dot{\theta}^2}{2} \frac{dr^2}{dr} + m\mu \frac{dr^{-1}}{dr} - \frac{3m\mu J_2 r_o^2}{2} \sin^2 \theta \frac{dr^{-3}}{dr} + \frac{m\mu J_2 r_o^2}{2} \frac{dr^{-3}}{dr} \\
&= mr\dot{\phi}^2 \cos^2 \theta + mr\dot{\theta}^2 - m\mu r^{-2} + \frac{9}{2}m\mu J_2 r_o^2 r^{-4} \sin^2 \theta - \frac{3}{2}m\mu J_2 r_o^2 r^{-4} \\
&= mr \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \mu r^{-3} + \frac{9}{2}\mu J_2 r_o^2 r^{-5} \sin^2 \theta - \frac{3}{2}\mu J_2 r_o^2 r^{-5} \right) \\
&= mr \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3\mu J_2 r_o^2}{2r^5} (3 \sin^2 \theta - 1) \right)
\end{aligned} \tag{45}$$

So, as per Eq. (36), we can then equate the above calculated terms to their respective non-conservative generalized drag force to get the equations of motion.

For the coordinate ϕ we set up the EoM as:

$$\begin{aligned}
&\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = Q_{\phi,nc} \\
&mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) - 0 = -\frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi}
\end{aligned} \tag{46}$$

And we then isolate the $\ddot{\phi}$ term:

$$\begin{aligned}
\ddot{\phi} m r^2 \cos^2 \theta &= -\frac{1}{2}C_D A \rho(r) v \dot{\phi} r^2 \cos^2 \theta - 2\dot{r}\dot{\phi} m r \cos^2 \theta + 2\dot{\phi}\dot{\theta} \sin \theta m r^2 \cos \theta \\
\ddot{\phi} &= -\frac{1}{2}C_D A \rho(r) v \dot{\phi} \frac{r^2 \cos^2 \theta}{m r^2 \cos^2 \theta} - 2\dot{r}\dot{\phi} \frac{m r \cos^2 \theta}{m r^2 \cos^2 \theta} + 2\dot{\phi}\dot{\theta} \frac{\sin \theta}{\cos \theta} \frac{m r^2 \cos \theta}{m r^2 \cos \theta} \\
\ddot{\phi} &= -\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\phi}\dot{\theta} \tan \theta
\end{aligned} \tag{47}$$

For the coordinate θ we set up the EoM as:

$$\begin{aligned}
&\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = Q_{\theta,nc} \\
&mr \left(2\dot{r}\dot{\theta} + r\ddot{\theta} \right) + m \cos \theta \sin \theta \left(r^2 \dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^3} \right) = -\frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta}
\end{aligned} \tag{48}$$

And we then isolate the $\ddot{\theta}$ term:

$$\begin{aligned}
\ddot{\theta} m r^2 &= -\frac{1}{2} C_D A \rho(r) v \dot{\theta} r^2 - 2 \dot{r} \dot{\theta} m r - \dot{\phi}^2 \cos \theta \sin \theta m r^2 - 3 \mu J_2 r_o^2 \cos \theta \sin \theta \frac{m}{r^3} \\
\ddot{\theta} &= -\frac{1}{2} C_D A \rho(r) v \dot{\theta} \frac{r^2}{m r^2} - 2 \dot{r} \dot{\theta} \frac{m r}{m r^2} - \dot{\phi}^2 \cos \theta \sin \theta \frac{m r^2}{m r^2} - 3 \mu J_2 r_o^2 \cos \theta \sin \theta \frac{m}{r^3} \frac{1}{m r^2} \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \dot{\phi}^2 \cos \theta \sin \theta - \frac{3 \mu J_2 r_o^2}{r^5} \cos \theta \sin \theta \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3 \mu J_2 r_o^2}{r^5} \right)
\end{aligned} \tag{49}$$

For the coordinate r we set up the EoM as:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} &= Q_{r,nc} \\
m \ddot{r} - m r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) &= -\frac{1}{2} C_D A \rho(r) v \dot{r}
\end{aligned} \tag{50}$$

And we then isolate the $\ddot{r}$ term:

$$\begin{aligned}
\ddot{r} m &= -\frac{1}{2} C_D A \rho(r) v \dot{r} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) m \\
\ddot{r} &= -\frac{1}{2} C_D A \rho(r) v \dot{r} \frac{1}{m} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) \frac{m}{m} \\
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) \\
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \dot{\phi}^2 \cos^2 \theta + r \dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3 \mu J_2 r_o^2}{2 r^4} (3 \sin^2 \theta - 1)
\end{aligned} \tag{51}$$

Which makes the EoM:

$$\begin{aligned}
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \dot{\phi}^2 \cos^2 \theta + r \dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3 \mu J_2 r_o^2}{2 r^4} (3 \sin^2 \theta - 1) \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3 \mu J_2 r_o^2}{r^5} \right) \\
\ddot{\phi} &= -\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2 \dot{r} \dot{\phi}}{r} + 2 \dot{\phi} \dot{\theta} \tan \theta
\end{aligned} \tag{52}$$

Where: $v = \sqrt{\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2}$

(b.4): Find

For zero drag, identify the cyclic coordinate and the associated conserved generalized momentum. Discuss its relation to the angular momentum about $\hat{e}_z$, denoted by H_z .

Analysis/Conclusions:

It was identified in (b.3) that the Lagrangian is not a function of ϕ . **This means that ϕ is the cyclic coordinate.**

If the force of drag is removed, this removes the only non-conservative force, and the Lagrange equation for ϕ now evaluates as:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = 0 \quad (53)$$

Because $\frac{\partial L}{\partial \phi} = 0$ as determined in (b.3), the generalized momentum p_ϕ is constant over time.

Likewise, the equation for this constant momentum was also found in (b.3) Eq. (37):

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2 \dot{\phi} \cos^2 \theta \quad (54)$$

Which can be evaluated from θ and $\dot{\phi}$ at any time.

Because the angle ϕ is the rotation of the spacecraft around the $\hat{e}_z$ axis, we can say that p_ϕ is **equivalent to the angular momentum about the $\hat{e}_z$ axis H_z .**

(b.5): Find

For zero drag, show that the Jacobi energy integral

$$\mathcal{J} := \dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}} - L \quad (55)$$

is identical to the total energy $E = T + V$. Discuss if $\mathcal{J}$ is conserved under no drag, and why.

Analysis/Conclusions:

The total energy E computes just like the Lagrangian in Eq. (30), but the potential is added instead of subtracted. Likewise, the expression ends up looking

quite similar:

$$\begin{aligned}
E &= T + V \\
&= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) - \frac{m\mu}{r} \left(1 - J_2 \left(\frac{r_o}{r} \right)^2 \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} - \frac{m\mu}{r} + \frac{m\mu J_2 r_o^2}{r^3} \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \quad (56) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} - \frac{m\mu}{r} + \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta - \frac{m\mu J_2 r_o^2}{2r^3}
\end{aligned}$$

Then to find the Jacobi integral, first we compute $\dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}}$. This is a dot product between the coordinate rates and the generalized momentum found in (b.3) in equations (37), (38), and (39):

$$\begin{aligned}
\dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}} &= \dot{r} \frac{\partial L}{\partial \dot{r}} + \dot{\theta} \frac{\partial L}{\partial \dot{\theta}} + \dot{\phi} \frac{\partial L}{\partial \dot{\phi}} \\
&= \dot{r} (m\dot{r}) + \dot{\theta} (mr^2 \dot{\theta}) + \dot{\phi} (mr^2 \dot{\phi} \cos^2 \theta) \quad (57) \\
&= m\dot{r}^2 + mr^2 \dot{\theta}^2 + mr^2 \cos^2 \theta \dot{\phi}^2
\end{aligned}$$

Then simply subtract the Lagrangian. We see below that this ends up flipping all the signs that were different between the Lagrangian and the total energy, and this ends up showing that the Jacobi integral is equal to the total energy:

$$\begin{aligned}
\mathcal{J} &= \dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}} - L \\
&= \left(m\dot{r}^2 + mr^2 \dot{\theta}^2 + mr^2 \cos^2 \theta \dot{\phi}^2 \right) \\
&\quad - \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} - \frac{m\mu}{r} + \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta - \frac{m\mu J_2 r_o^2}{2r^3} = E \quad (58)
\end{aligned}$$

In our zero drag system, **the Jacobi integral is conserved**. This is because the Jacobi integral represents generalized total energy, and for it to be conserved it must not change over time. It will change over time if: there are non-conservative forces breaking conservation of energy in the system, or there is time variance in the Lagrangian, since $\mathcal{J}$ is purely a function of L . There are no non-conservative forces, and we see from (b.3) Eq. (30) that L has no explicit dependence on time so $\mathcal{J}$ is conserved.

(b.6): Find

Under drag, derive the time derivative of H_z and $\mathcal{J}$. Discuss if they are conserved and why.

Analysis/Conclusions:

As discussed in (b.4), $H_z = p_\phi = \frac{\partial L}{\partial \dot{\phi}}$, thus the time derivative $\frac{d}{dt}(\frac{\partial L}{\partial \dot{\phi}})$ found in Eq. (40) is equivalent to the time derivative of H_z :

$$\frac{d}{dt}H_z = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) \quad (59)$$

The time rate of the Jacobi integral $\mathcal{J}$ is given by:

$$\frac{d\mathcal{J}}{dt} = \mathbf{Q}_{\text{nc}} \cdot \dot{\mathbf{q}} - \frac{\partial L}{\partial t} \quad (60)$$

The dot product portion of Eq. (60) is between non-conservative forces found in Eq. (35) and the generalized coordinate rate, and it evaluates to:

$$\begin{aligned} \mathbf{Q}_{\text{nc}} \cdot \dot{\mathbf{q}} &= [Q_r \quad Q_\phi \quad Q_\theta] \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \end{bmatrix} \\ &= \left[-\frac{1}{2}C_D A \rho(r) v \dot{r} \quad -\frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \quad -\frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta} \right] \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \end{bmatrix} \\ &= -\frac{1}{2}C_D A \rho(r) v \dot{r}^2 - \frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi}^2 - \frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta}^2 \\ &= -\frac{1}{2}C_D A \rho(r) v \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) \\ &= -\frac{1}{2}C_D A \rho(r) v^3 \end{aligned} \quad (61)$$

The portion in the parenthesis ends up being equivalent to $\|\mathbf{v}\|_2^2$, so it simplifies to a fairly compact scalar function.

Looking at the Lagrangian L :

$$L = \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \quad (62)$$

As discussed in (b.5), there are no explicit time t terms, so the partial $\frac{\partial L}{\partial t} = 0$.

Which makes the time derivative of $\mathcal{J}$ equal to:

$$\frac{d\mathcal{J}}{dt} = \mathbf{Q}_{\text{nc}} \cdot \dot{\mathbf{q}} - \frac{\partial L}{\partial t} = -\frac{1}{2}C_D A \rho(r) v^3 - 0 = -\frac{1}{2}C_D A \rho(r) v^3 \quad (63)$$

So, to reiterate, the time derivatives of H_z and $\mathcal{J}$ are:

$$\begin{aligned}\frac{d}{dt}H_z &= mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) \\ \frac{d}{dt}\mathcal{J} &= -\frac{1}{2}C_D A \rho(r) v^3\end{aligned}\tag{64}$$

We see that under drag H_z is not conserved because... idk

Then, we see that $\mathcal{J}$ is not conserved because its rate of change is a constant decrease scaling with position and velocity. Since neither term can be negative themselves, the sign never flips and energy always decreases under drag.